

P3 - Paneling - Descriptive Geometry and Arrays

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Revision: 2026-09-22. Dates and weights follow the course schedule.

Current scope: descriptive geometry and 1D, 2D and 3D arrays. Attractors and other array families are reserved for later work.

P3a - Descriptive geometry module - material and lighting

Introduced: Week 4 - Sep 15 / **Due:** Week 6 - Sep 29 / **Weight:** 7%

Description

Construct a panel from a built precedent using explicit point positions and ordered faces. Explain how its parameters determine the geometry, verify that the panel closes and tiles, and present it with a deliberate material and lighting setup. This authored module becomes the unchanged element in P3b.

Rules & Constraints

- Choose a simple panel from a built precedent that tiles edge to edge in a 2 x 2 grid. Credit the building, architect and source.
- Define parameters first and explain each. Include a shape control such as aperture, fold depth or corner offset, beyond overall dimensions.
- Build an editable, closed mesh from explicit point and face lists. Calculate point positions from the parameters.
- State the ranges where the module closes and tiles. Diagram low, middle and high values of a shape parameter.
- Use one 17 x 11 inch sheet with one studio render of the module on a ground plane - one material, a deliberate lighting setup (key, fill and rim, or one sun and sky) and no entourage. Record the material and light settings on the sheet.
- Beside the render, show point and face lists, a parameter table and the vertex order of one face. Explain normals, closure and reversed face winding.

Parameters

- Width, depth and height/thickness in stated model units, with a local origin and X/Y/Z axes.
- At least one shape control beyond overall size: aperture, fold depth, corner offset or an equivalent geometric feature. Explain how this value changes specific point coordinates.
- Valid ranges and dependencies for every control. State the limits within which faces remain valid, the solid closes and neighboring panels meet.
- Material settings and light positions/intensities relevant to the studio render. Use one material and either key/fill/rim lighting or one sun with sky.

Required Studies

- **Precedent and construction.** Credit the building, architect and image source. Diagram the panel boundary, numbered vertices, connecting edges and ordered face lists. Show the vertex order of one face and explain how reversing it changes its normal.
- **Descriptive views.** Use coordinated plan/front/side views and an axonometric construction diagram to show how the point coordinates generate the form. Dimension the boundary and the shape-controlled feature; distinguish construction lines from final edges.

- **Three parameter settings.** Show low, middle and high values of one shape parameter from the same graph. Label all settings. Verify closure and show how the selected module tiles edge to edge in a 2 x 2 grid.
- **One studio render.** Present the module on a ground plane, with one material and intentional lighting, without entourage. Record the material and lighting settings beside the image so the result can be reproduced.

Workflow

1. Select a simple built panel with a repeatable boundary. Identify its point/edge/face relationships before creating the graph.
2. Define named inputs, calculate the vertices and assemble the face lists in consistent order. Build the module yourself from these elements.
3. Check closure, normals and shared tile boundaries at the low/middle/high parameter settings. Resolve gaps, reversed faces or self-intersections.
4. Set up the material, ground and lights. Compose one sheet that explains both the geometric construction and its rendered appearance.

Deliverables

- **One 17 x 11 inch landscape module sheet:** Lastname_Firstname_P3a.pdf. Include the credited precedent, construction and descriptive views, point/face lists, parameter table, three shape settings, 2 x 2 tiling check and studio render with settings.
- **Editable graph/source:** .blend, or an already approved .gh/.3dm route. Include the material/light scene used for the render and package referenced assets. Keep native node names visible and explain your groups with Frames.
- Submit files in Lastname_Firstname_P3a/. Save the chosen parameter setting and panel origin so the exact same module can be reused in P3b. No physical print is required for this milestone.

Evaluation

- **Geometric reasoning:** point and face lists, ordered vertices, normals and closure explain the constructed solid.
- **Control and validity:** a meaningful shape input produces the three documented alternatives inside a tested range.
- **Tiling:** the selected panel meets its neighbors edge to edge in the 2 x 2 check.
- **Presentation:** drawings agree with the render; material, lighting and graphic hierarchy reveal the panel geometry.

Before You Submit

- [] Built precedent credited; panel authored from calculated points and ordered faces.
- [] One shape control beyond dimensions, tested at low/middle/high values.
- [] Closure, normals and edge-to-edge 2 x 2 tiling demonstrated.
- [] Point/face lists, descriptive views and parameter values are readable.
- [] One material, deliberate lights, ground plane and no entourage; source and sheet match.

P3b - Arrays

Introduced: Week 6 - Sep 29 / **Due:** Week 8 - Oct 13 / **Weight:** 7%

Description

Use your P3a panel to explain repetition in one, two and three dimensions. Keep the module geometry unchanged and compute its placement from indices, counts and spacing. Show both the arrangement and the rule that generates it. Attractors and other array families are outside this brief's current scope.

Rules & Constraints

- Array your P3a module across exactly two 17 x 11 inch sheets, submitted together as one assignment.
- Sheet 1 - 1D, 2D and 3D arrays: a linear array, a nested grid and an XYZ cube array. Label count, spacing and index order; show the predicted total for each.
- Sheet 2 - parameters and index logic: a second count/spacing setting for each 1D, 2D and 3D array, a row/column/layer diagram, and the conditional selection before and after its rule applies.
- Current brief scope: 1D, 2D and 3D arrays. Hexagonal, radial and curve arrays and attractors are reserved for later work.
- Keep the module unchanged; vary only its placement. Expose count and spacing or radius in the graph.
- Build repetition from basic nodes and begin with a count you can inspect; there is no minimum element count.
- Include one conditional selection on either sheet. State its rule (for example, indices 3-5 lift by 0.5) and show the array before and after it applies.

Parameters

- Integer counts N_x , N_y , N_z and positive step distances dx , dy , dz . Step distance measures between corresponding module origins; it is not necessarily the clear gap between edges.
- Base point P_0 , array axes and one shared module orientation. Label the local origin used to place each copy.
- A conditional selection rule and its effect, such as lifting a named index range. Expose the selection limits and displacement without changing the module itself.

Required Studies

- **1D - row.** Vary one index: $P(i) = P_0 + i \cdot dx \cdot X$. Label the first few indices, count and spacing. Predict the number of copies before evaluating the graph.
- **2D - grid.** Use independent row/column indices: $P(i,j) = P_0 + i \cdot dx \cdot X + j \cdot dy \cdot Y$. Predict $N_x \cdot N_y$ copies. Show the row/column structure and distinguish it from manually duplicating a row.
- **3D - spatial array.** Add an independent layer index: $P(i,j,k) = P_0 + i \cdot dx \cdot X + j \cdot dy \cdot Y + k \cdot dz \cdot Z$. Predict $N_x \cdot N_y \cdot N_z$ copies. For equal counts N , the total is N cubed; 3, 4 and 5 per axis give 27, 64 and 125 copies. Show an axonometric view plus a layer view that makes the Z repetition visible.
- **Controlled comparisons.** For each array, change count and/or spacing using the same graph and record both settings and totals. Keep the panel shape fixed. Begin with inspectable counts; no large minimum element count is required.
- **One conditional.** State an index-based rule and show before/after views. Identify the affected copies and the displacement. Ensure the selection leaves the other copies unchanged.

Workflow

1. Reference the selected P3a module, preserving its dimensions, shape controls and origin. Separate panel generation from placement.
2. Build a small 1D row, then nested 2D and 3D repetitions from basic nodes. Use $i = 0..N_x-1$, $j = 0..N_y-1$, $k = 0..N_z-1$ and define X/Y/Z as unit vectors of the stated axes.
3. Predict and inspect counts at each stage. Explain indices and row/layer organization. Distinguish Grasshopper lists/trees from Blender fields or Repeat Zones; use the appropriate data structure for your software.
4. Test the second count/spacing settings and the conditional selection. Check for duplicate positions and unintended collisions, then compose the two sheets.

Deliverables

- **Exactly two 17 x 11 inch landscape sheets in one PDF:** Lastname_Firstname_P3b.pdf. They are one assignment, submitted together.
- **Sheet 1 - 1D, 2D and 3D.** Show the row, nested grid and spatial array with their placement rules, labeled axes, count/spacing values and predicted totals. Include the 3D layer view and a small reference image of the unchanged P3a panel.
- **Sheet 2 - Parameters and index logic.** Show the second setting for each array with values and totals, a diagram of row/column/layer organization, and the conditional before/after comparison with its rule.
- **Editable graph/source** in Lastname_Firstname_P3b/, with labeled inputs and an output for each array. Internalize Grasshopper references or package equivalent dependencies. Save/record the settings used on both sheets so they can be reproduced. No physical print is required.

Evaluation

- **Array logic:** row, grid and spatial array use one, two and three independent directions; predicted and actual totals agree.
- **Data control:** index order and row/layer organization are understandable; the conditional affects exactly the intended copies.
- **Parametric behavior:** count and spacing changes recompute the arrangement while preserving the P3a module.
- **Design and communication:** views reveal spacing and dimensional structure; captions, diagrams and file organization follow the Graphic Standards Manual.

Before You Submit

- [] Same P3a panel used unchanged in all arrangements.
- [] A row, a grid and a true spatial array are clearly distinguished.
- [] Axes, count, spacing, index order and predicted totals are shown for each.
- [] A second setting for every array and one conditional before/after comparison are documented.
- [] Two sheets submitted together; the source reopens and reproduces the documented outputs.

Shared standards: Course software, fabrication, attribution and AI policies apply. Submit through the channel announced in class/Canvas. Test the packaged source from a clean folder. Percentages are existing course weights. This handout uses US Letter; student sheets are 17 x 11 inches.